

Matrix Multiplication (Optional)

This section explains how matrix multiplication works, building up from the simpler vector dot product. This is the core mathematical operation that makes vectorization possible.

1. Vector-Vector Dot Product

The **dot product** is a fundamental operation where you take two vectors of the same size, multiply their corresponding elements, and sum the results.

- **Example:**
 - $\vec{a} = [1, 2]$
 - $\vec{w} = [3, 4]$
 - **Dot Product:** $z = (1 \times 3) + (2 \times 4) = 3 + 8 = 11$
 - **General Formula:** $z = a_1w_1 + a_2w_2 + \dots + a_nw_n$
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2. The Transpose and the "Row $\times$ Column" Rule

- A **column vector** is a matrix with one column (e.g., a 2x1 matrix).
 - A **row vector** is a matrix with one row (e.g., a 1x2 matrix).
 - The **transpose** operation (denoted by A^T) converts a column vector into a row vector (it lays the vector "on its side").
 - **Key Equivalence:** The vector dot product $\vec{a} \cdot \vec{w}$ is mathematically equivalent to the matrix multiplication of $\vec{a}^T$ (a row vector) and $\vec{w}$ (a column vector).
 - $z = \vec{a} \cdot \vec{w} \equiv z = \vec{a}^T \times \vec{w}$
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3. Vector-Matrix Multiplication

This is the process of multiplying a row vector by a matrix.

- **Rule:** You perform a dot product between the **row vector** and **each column** of the matrix. The results form a new row vector.
- **Example:**
 - $\vec{a}^T = [1, 2]$ (a 1x2 matrix)
 - $W = \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$ (a 2x2 matrix, with columns $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 6 \end{bmatrix}$)
- **Calculation:**
 1. **First element of output:** $(\vec{a}^T) \cdot (\text{Col 1 of } W)$

- $[1, 2] \cdot [3, 4] = (1 \times 3) + (2 \times 4) = 11$

2. **Second element of output:** $(\vec{a}^T) \cdot (\text{Col 2 of } W)$

- $[1, 2] \cdot [5, 6] = (1 \times 5) + (2 \times 6) = 17$

- **Result:** $Z = [11, 17]$ (a 1x2 matrix)

4. Matrix-Matrix Multiplication

This is the general form. It simply repeats the vector-matrix multiplication process for **every row** of the first matrix.

- **Rule:** The element at (row i , col j) of the output matrix is the dot product of (row i) of the *first* matrix and (col j) of the *second* matrix.

- **Example:** $Z = A^T \times W$

- $A^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$

- $W = \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$

- **Calculation:**

1. **Row 1 of Z:** (Row 1 of A^T) $\times$ W

- $([1, 2] \cdot [3, 4]) = 11$

- $([1, 2] \cdot [5, 6]) = 17$

- $\rightarrow$ First row of Z is $[11, 17]$

2. **Row 2 of Z:** (Row 2 of A^T) $\times$ W

- $([-1, -2] \cdot [3, 4]) = (-3) + (-8) = -11$

- $([-1, -2] \cdot [5, 6]) = (-5) + (-12) = -17$

- $\rightarrow$ Second row of Z is $[-11, -17]$

- **Final Result:** $Z = \begin{bmatrix} 11 & 17 \\ -11 & -17 \end{bmatrix}$